

Chemistry, 2017 EC, natural science, entrance exam

Grade 12 · Chemistry · 2017 E.C.

Time allowed: 120 minutes

Questions: 43

Total marks: 43

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following CORRECTLY shows the relative charges of an electron, proton, and neutron, respectively? (1)

A. 0, +1, -1

C. -1, +1, 0

B. 0, -1, +1

D. -1, 0, +1

2. Which statement below CORRECTLY describes Bohr's model of an atom? (1)

A. Protons and electrons are found in the nucleus.

B. Electrons and neutrons are found in the nucleus.

C. Protons move in circular orbits around the nucleus.

D. Electrons move in circular orbits around the nucleus.

3. Which statement below CORRECTLY describes periodicity? Periodicity is (1)

A. a regular repetition of chemical and physical properties in the Periodic Table.

B. a row in the Periodic Table which contains elements with the same number of shells.

C. a measure of electronegativity which runs from least to most electronegative elements.

D. a column of the Periodic Table which contains elements with similar chemical properties.

4. Which statement below CORRECTLY expresses hydrogen bonding? It is formed by bonding (1)

A. a hydrogen atom to an element such as sodium.

B. a hydrogen atom to an element such as oxygen.

C. two hydrogen atoms with equal sharing of electrons.

D. two hydrogen atoms with unequal sharing of electrons.

5. What is the bond formed between two or more atoms by sharing of electrons? (1)

A. Ionic bond

C. Covalent bond

B. Metallic bond

D. Valence bond

6. Which statement below CORRECTLY describes dipole-dipole forces? It is the force that exists (1)

A. in all molecules

C. between non-polar molecules.

B. between polar molecules.

D. between polar and non-polar molecules.

7. Comparing the forces between molecules of similar molar masses, which one of the following produces the weakest force? (1)

A. Covalent bonds.

C. Ionic forces.

B. Hydrogen bonding.

D. Van der Waal's forces.

8. What is a limiting reactant? It is the reactant that (1)

A. is regenerated at the end of the reaction.

B. remains unreacted as the reaction proceeds.

C. is present in excess when the reaction goes to completion.

D. is completely consumed when the reaction goes to completion.

9. Which statement below CORRECTLY distinguishes redox and non-redox reactions? (1)

 - A. Electrons are transferred in redox reactions but not in non-redox reactions.
 - B. Reduction occurs in redox reactions and oxidation takes place in non-redox reactions.
 - C. Oxidation number remains the same in both redox and non-redox reactions.
 - D. Acid-base reactions can be considered as both redox and non-redox reactions.

10. Given the reaction $\text{C(s)} + \text{CO}_2\text{(g)} \rightleftharpoons 2\text{CO(g)}$, what is the equilibrium constant expression? (1)

 - A. $[\text{CO}]^2 / [\text{C}][\text{CO}_2]$
 - B. $[\text{CO}]^2 / [\text{CO}_2]$
 - C. $[\text{CO}]^2 / [\text{C}]$
 - D. $[\text{C}][\text{CO}_2] / [\text{CO}]^2$

11. Which of the following pair of substances can be considered as examples of the liquid state of matter at room temperature and 1 atm pressure? (1)

 - A. Carbon and mercury
 - B. Water and carbon dioxide
 - C. Water and mercury
 - D. Carbon and carbon monoxide

12. Which gas law describes the behavior of gases using the variables: temperature, volume, pressure and number of moles? (1)

 - A. Ideal gas law
 - B. Boyle's law
 - C. Avogadro's law
 - D. Combined gas law

13. Which statement below describes CORRECTLY the assumptions in the kinetic theory of gases? (1)

 - A. Gas molecules occupy a finite volume and have indefinite shape.
 - B. Gas molecules are in random motion with no interactions.
 - C. The average kinetic energies of gas molecules are independent of temperature.
 - D. The average kinetic energies of gas molecules are independent of the amount of gas.

14. If the rate of diffusion of a certain unknown gas is half times the rate of diffusion of helium, what will be the molar mass of the unknown gas? (Atomic mass of He = 4.0 g/mole) (1)

 - A. 2.0 g/mole
 - B. 8.0 g/mole
 - C. 16.0 g/mole
 - D. 32.0 g/mole

15. A student added boiling chips into a beaker of water while carrying out an activity to determine the boiling point of water. Why do you think the boiling chips are necessary? (1)

 - A. To ensure smooth rate of evaporation
 - B. To speed up the rate of evaporation
 - C. To maintain constant temperature and pressure
 - D. To reach the boiling point with less consumption of heat

16. What is the oxidation number of manganese in KMnO_4 ? (Atomic Number K=19, Mn=25, O=16) (1)

 - A. +5
 - B. +6
 - C. +7
 - D. +8

17. Aluminum reacts with oxygen to form aluminum oxide. What is the coefficient of aluminum after balancing the reaction? (Atomic number of Al=13, O=16) (1)

 - A. 2
 - B. 4
 - C. 6
 - D. 8

18. What property of an ionic compound can be investigated by heating it in a crucible? (1)

 - A. Melting point
 - B. Solubility
 - C. Malleability
 - D. Crystallization point

19. Silver has two known isotopes, ^{107}Ag and ^{109}Ag with their percent abundance of 52% and 48% respectively. What is the average atomic mass of silver? (1)
- A. 107.89
B. 108.06
C. 107.48
D. 108.86
20. Which reaction scheme below illustrates the formation of a cation? (1)
- A. $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$
B. $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
C. $\text{S} + 2\text{e}^- \rightarrow \text{S}^{2-}$
D. $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$
21. Which law governs the fact that the ratio of the amount of hydrogen to oxygen by weight will be the same in a drop of water and a lake of water? (1)
- A. Law of relative composition
B. Law of conservation of mass
C. Law of multiple proportions
D. Law of definite composition
22. Which statement below CORRECTLY describes the characteristics of chemical equilibrium? (1)
- A. A reaction stops when it reaches equilibrium
B. A reaction stops if sufficient amount of time is given.
C. The rates of the forward and reverse reactions are equal.
D. A new substance or substances are formed when equilibrium is attained.
23. What is the CORRECT IUPAC name of this alkane? (1)
-  Question Diagram
- A. 5,7-dichloro-3,9-dimethyldecane
B. 4,6-dichloro-2,8-dimethyldecane
C. 4,6-dichloro-2-ethyl-8-methylnonane
D. 4,6-dichloro-8-ethyl-2-methylnonane
24. What is the name given to compounds of carbon and hydrogen containing multiple bonds? (1)
- A. Aliphatic hydrocarbons
B. Aromatic hydrocarbons
C. Saturated hydrocarbons
D. Unsaturated hydrocarbons
25. Which organic compound is used for producing 1,2-ethanediol? (1)
- A. Ethene
B. Ethane
C. Ethyne
D. Ethanol
26. Which of the following is the main constituent of natural gas? (1)
- A. Ethane
B. Butane
C. Propane
D. Methane
27. In order to avoid the decrease in quality of some clothes after washing with water, which of the following chemicals is used for dry cleaning? (1)
- A. CH_4
B. CH_2Cl_2
C. C_2H_4
D. C_2Cl_4
28. Which of the following properties is the property of an acidic oxide? An acidic oxide reacts with (1)
- A. salts to form acids.
B. water to form acids.
C. bases to form basic hydroxide and water.
D. basic oxides to form hydroxides and water

29. What is the name given to an acid that dissociates (ionizes) only to a slight extent in aqueous solution? (1)
- A. Dilute acid
B. Strong acid
C. Weak acid
D. Concentrated acid
30. Solutions can be classified based upon their pH values. What is a solution whose pH value 13? (1)
- A. Amphoteric solution
B. Acidic solution
C. Neutral solution
D. Basic solution
31. During laboratory class, a student placed clean water in a beaker and added pieces of calcium metal using tongs. After the reaction is completed, he/she tested the solution using red litmus paper. Which of the following will be observed? (1)
- A. The litmus paper turns to blue.
B. The litmus paper turns to red.
C. The litmus paper maintains its color.
D. The litmus paper turns to colorless.
32. In an activity during laboratory session a small amount of lead bromide (PbBr_2) crystals were placed in a beaker. Two electrodes are inserted until they are in contact with the PbBr_2 crystals. When the PbBr_2 crystals in the beaker are gently heated, which of the following will be observed? (1)
- A. Bromine gas is evolved at the anode
B. Bromide ion is oxidized at the cathode
C. Lead (II) ion is reduced at the anode
D. Lead (II) ion is oxidized at the cathode
33. What is the name given to the pollution caused by the dumping of non-biodegradable wastes into the environment? (1)
- A. Air pollution
B. Land pollution
C. Sound pollution
D. Water pollution
34. What is the purpose of the conversion of pig iron to steel? (1)
- A. To remove impurities by oxidation
B. To remove impurities by reduction
C. To increase the concentration of iron in the pig iron
D. To decrease the concentration of iron in the pig iron
35. Which of the following statements CORRECTLY describes the law of conservation of mass? (1)
- A. The total mass of substances varies during a chemical reaction.
B. Mass is neither created nor destroyed during a chemical reaction.
C. A particular compound is composed of the same elements in the same parts by mass.
D. Pure compounds always contain the elements in the same percentage by mass.
36. A mineral absorbs purple light of frequency 7.11×10^{14} Hz. What is the wavelength (in nm) of the absorbed light? (1)
Given: $c = 3.00 \times 10^8 \text{ m/s}$
- A. 184.4 nm
B. 237.5 nm
C. 421.9 nm
D. 514.5 nm
37. Why do atoms absorb energy when their electrons undergo transitions from a lower energy to a higher energy level? (1)
This is because,
- A. the electrons of the atoms move from one orbit of lower radius to another one having larger radius.
B. the electrons of the atoms move from one orbit of higher radius to another one having lower radius.
C. the electrons of the atoms do not have allowable energy levels.
D. electrons of the atoms absorb energy when they go from higher to lower energy.

38. What are the quantum numbers for the 5s orbital? (1)

- A. $n = 5, \quad l = 0, \quad m_l = -1, \quad m_s = \pm \frac{1}{2}$
- B. $n = 5, \quad l = 1, \quad m_l = -2, \quad m_s = \pm \frac{1}{2}$
- C. $n = 5, \quad l = 2, \quad m_l = -2, \quad m_s = \pm \frac{1}{2}$
- D. $n = 5, \quad l = 0, \quad m_l = 0, \quad m_s = \pm \frac{1}{2}$

39. Which is CORRECT about the quantum numbers of helium's two electrons? (1)

- A. Both electrons have the same four quantum numbers.
- B. The second electron occupies the same orbital as the first with the same spin.
- C. The second electron occupies the same orbital as the first with opposite spin.
- D. The magnetic quantum numbers of both electrons are different.

40. Scandium (Sc) has the configuration [Ar] 4s² 3d¹. To which group does it belong? (1)

- A. Representative elements
- B. Transition elements
- C. Inner transition elements
- D. Nonmetals

41. A student carried out an experiment to determine the melting points of NaCl and CuCl₂. He/she added 0.2 g of NaCl and CuCl₂ into two different test tubes and heated both tubes simultaneously using a Bunsen burner. Which of the following statements is CORRECT regarding the melting point of NaCl and CuCl₂? (1)

- A. The melting points of NaCl are lower than that of CuCl₂.
- B. The melting point of NaCl is higher than that of CuCl₂.
- C. Both NaCl and CuCl₂ will start to melt at the same time.
- D. Both NaCl and CuCl₂ will not melt.

42. What is the cause of unusual high boiling points of HF, H₂O and NH₃? (1)

- A. Dipole-Dipole forces
- B. Induced dipole forces
- C. Hydrogen bonding
- D. Ion-dipole forces

43. What is the electron configuration of O₂ molecule as per the molecular orbital model? (Atomic number of O = 8) (1)

- A. $\sigma_{1s}^2 \quad \sigma_{1s}^{*2} \quad \sigma_{2s}^2 \quad \sigma_{2s}^{*2} \quad \sigma_{2p_x}^2 \quad \pi_{2p_y}^2 = \pi_{2p_z}^2 \quad \pi_{2p_y}^{*1} = \pi_{2p_z}^{*1}$
- B. $\sigma_{1s}^2 \quad \sigma_{1s}^{*2} \quad \sigma_{2s}^2 \quad \sigma_{2s}^{*2} \quad \sigma_{2p_x}^2 \quad \pi_{2p_y}^2 = \pi_{2p_z}^2 \quad \pi_{2p_y}^{*2} = \pi_{2p_z}^{*2}$
- C. $\sigma_{1s}^2 \quad \sigma_{1s}^{*2} \quad \sigma_{2s}^2 \quad \sigma_{2s}^{*2} \quad \sigma_{2p_x}^2 \quad \pi_{2p_y}^2 = \pi_{2p_z}^2 \quad \pi_{2p_y}^{*2} = \pi_{2p_z}^{*2}$
- D. $\sigma_{1s}^2 \quad \sigma_{1s}^{*2} \quad \sigma_{2s}^2 \quad \sigma_{2s}^{*2} \quad \sigma_{2p_x}^2 \quad \pi_{2p_y}^2 = \pi_{2p_z}^2 \quad \pi_{2p_y}^{*3} = \pi_{2p_z}^{*3}$

— END OF EXAMINATION —